



Image Processing

Lab 5

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Imports

```
In [1]: import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Read Image

```
In [9]: # Read the image
image = cv2.imread('images/Parrot.png')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Display the original image
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

The history saving thread hit an unexpected error (OperationalError('attempt to write a readonly database')).History will not be written to the database.

Original Image



Task 1

7×7 Box Filter

```
In [10... # Apply 7x7 box filter
box_filtered = cv2.blur(image, (7, 7))

# Plot original and smoothed image side by side
fig, ax = plt.subplots(1, 2, figsize=(12, 6))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(box_filtered)
ax[1].set_title("Smoothed Image (7x7 Box Filter)")
ax[1].axis("off")

plt.show()
```

Original Image



Smoothed Image (7x7 Box Filter)



Task 2

Gaussian Filters (5x5 and 21x21)

```
In [12... # Apply 5x5 Gaussian filter
gaussian_5x5 = cv2.GaussianBlur(image, (5, 5), 0)

# Apply 21x21 Gaussian filter
gaussian_21x21 = cv2.GaussianBlur(image, (21, 21), 0)

# Plot the three images side by side
fig, ax = plt.subplots(1, 3, figsize=(18, 6))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(gaussian_5x5)
ax[1].set_title("5x5 Gaussian Filter")
ax[1].axis("off")

ax[2].imshow(gaussian_21x21)
ax[2].set_title("21x21 Gaussian Filter")
ax[2].axis("off")

plt.show()
```

Original Image



5x5 Gaussian Filter



21x21 Gaussian Filter



Task 3

3×3 Laplacian Sharpening

```
In [13... # Convert image to float32
image_float = image.astype(np.float32) / 255.0

# Apply Laplacian filter
laplacian = cv2.Laplacian(image_float, cv2.CV_32F, ksize=3)

# Sharpen the image
sharpened = np.clip(image_float - laplacian, 0, 1)

# Plot original, Laplacian result, and sharpened image
fig, ax = plt.subplots(1, 3, figsize=(18, 6))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(np.abs(laplacian), cmap='gray')
ax[1].set_title("Laplacian Filter Result")
ax[1].axis("off")

ax[2].imshow(sharpened)
ax[2].set_title("Sharpened Image")
ax[2].axis("off")

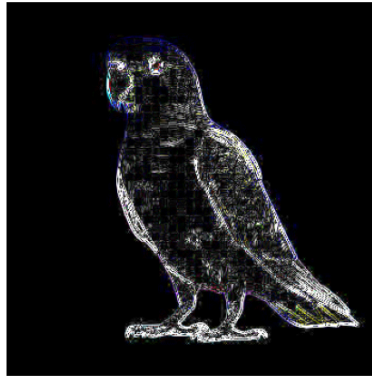
plt.show()
```

Clipping input data to the valid range for imshow with RGB data ([0..1] f or floats or [0..255] for integers). Got range [0.0..4.1254897].

Original Image



Laplacian Filter Result



Sharpened Image



In []: